

Microlensing and Difference Imaging Analysis

Third Year Lab Report

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Microlensing refers to gravitational lensing by compact objects such as stars or planets, observed as a time-varying magnification of the source flux. In this experiment, data from the Andromeda Galaxy Stellar Robotic Microlensing (Angstrom) Project were analysed using difference imaging analysis (DIA) with the *Isis* package to identify microlensing candidates in the M31 bulge and estimate the masses of compact objects. The results show that both substellar and stellar-mass lenses can produce detectable microlensing signals in the M31 bulge.

1. INTRODUCTION

Microlensing refers to gravitational lensing by compact objects such as stars or planets, where the angular separation of the lensed images cannot be resolved [1]. Its effect is instead observed by a time-varying magnification of the source flux. Since microlensing is achromatic and primarily depends on the lens mass, it can be used to infer lens masses. In 1986, Paczyński proposed to search for Massive Compact Halo Objects (MACHOs) as dark matter candidates [2].

The Andromeda Galaxy Stellar Robotic Microlensing Project (Angstrom Project) used stellar microlensing in M31 to study bulge structure and the stellar mass function [3]. In this experiment, Angstrom survey data are analysed with difference imaging analysis (DIA) to identify microlensing candidates and estimate the masses of compact objects in M31 galactic bulge.

2. THEORY

2.1. Gravitational Lensing

In GR, mass-energy warps spacetime and deflects massless particles such as light, so compact mass can act as a gravitational lens. In the point-source, point-lens (PSPL) modelling, perfect alignment of source, lens and observer produces an Einstein ring of angular radius

$$\theta_E = \sqrt{\frac{4GM}{c^2} \frac{D_S - D_L}{D_L D_S}} \quad (1)$$

where G is the gravitational constant, where M is the lens mass, D_S is the observer–source distance and D_L is the observer–lens distance.

It is convenient to define the impact parameter u as the source–lens angular separation in the units of θ_E . For rectilinear motion,

$$u(t) = \sqrt{u_0^2 + \left(\frac{t - t_0}{t_E}\right)^2} \quad (2)$$

where u_0 is the minimum impact parameter at time t_0 and t_E is the time required for source to cross the Einstein radius. When the lensed images are unresolved, the observable is the magnification

$$A = \frac{u^2 + 2}{u\sqrt{u^2 + 4}} \quad (3)$$

which relates the lensed and unlensed fluxes. This formalism follows Ref. [2].

2.2. Pixel Lensing Modelling

In the pixel lensing regime, where many unlensed sources are blended with the lensed source and magnification exceeds 10, the flux variation at a single point can be modelled by

$$F(t) = F_{\text{base}} + \frac{\Delta F_0}{\sqrt{1 + 12 \left(\frac{t - t_0}{t_{1/2}}\right)^2}} \quad (4)$$

where $F(t)$ is the observed flux, F_{base} is the baseline flux, ΔF_0 is the peak flux excess and $t_{1/2}$ is the full width half max (FWHM) of the microlensing light curve. In this regime, it is no longer able to determine the source flux, peak magnification and t_E independently.

The lens mass can be written, from Equation (1), as

$$M = \frac{c^2 D_S D_L}{4G(D_S - D_L)} \left(\frac{v_{\text{rel}} t_E}{D_L}\right)^2 \quad (5)$$

where v_{rel} is the relative transverse speed on the lens plane. Using Equation (4) with the high magnification approximation, t_E can be determined as

$$t_E = \frac{t_{1/2} A_0}{2\sqrt{3}} \quad (6)$$

where A_0 is a maximum magnification at u_0 . Thus, estimating M requires prior assumptions about A_0 , v_{rel} , D_S and D_L . This section follows Ref. [4].

2.3. Difference Imaging

Photometric time series are influenced by varying seeing, airmass and sky background, which can mimic intrinsic variability. Difference imaging, first introduced by Alard and Lupton, addresses this by isolating the true flux variation that is independent of these observational effects [5].

To extract the “true” flux change, one might attempt to select a base image and subtract the target image from it. However, the base and target images generally have different image profiles and simple subtraction does not yield ideal results. The discrepancies arise from two main sources: difference in the point spread function (PSF), where individual point sources appear with different shapes, and overall background flux differences.

To correct these effect, the base image can be modelled as

$$I(x, y) = R(x, y) \otimes K(x, y) + B(x, y) \quad (7)$$

where R is a high-quality reference image, K is a convolutional kernel that transforms PSF of R to that of the target images, B represents the background function and (x, y) are pixel coordinates. The kernel is expanded as a sum of Gaussian basis functions multiplied by low-order polynomials,

$$K(x, y) = \sum_{n=1}^{\mathcal{K}} \sum_{\substack{i_n, j_n \geq 0 \\ i_n + j_n \leq N_n}} a_{n, i_n, j_n} \exp\left(-\frac{x^2 + y^2}{2\sigma_n^2}\right) x^{i_n} y^{j_n}. \quad (8)$$

where $\mathcal{K}$ is the number of Gaussian components, N_n is the polynomial degree for component n , and σ_n is its width. Similarly, The background, B can be modelled as

$$B(x, y) = \sum_{\substack{\ell, m \geq 0 \\ \ell + m \leq N_B}} a_{\ell, m} x^\ell y^m \quad (9)$$

where N_B is the degree of the polynomial for the background function.

For given set of $(N_B, \mathcal{K}, \{N_n\}, \{\sigma_n\})$, the coefficients a of each term are obtained by minimising

$$\int D^2(x, y) dx dy = \int [I(x, y) - T(x, y)]^2 dx dy, \quad (10)$$

yielding an optimised difference image $D(x, y) = I(x, y) - T(x, y)$ from which light curves at fixed position can be extracted.

3. EXPERIMENTAL APPROACH

In this experiment, DIA was performed using the *Isis* package [6]. Prior to generating the full set of difference images, a reference image and the kernel/background free parameters $N_B, \mathcal{K}, \{N_n\}, \{\sigma_n\}$ were determined.

3.1. Building Reference Image

The reference image was constructed by stacking multiple high-quality images to maximise signal-to-noise ratio. Images were ranked using a score Q based on their metadata (seeing, airmass, background level and background RMS). For each quantity, a normalised score f_i between 0 and 1 was assigned, peaking near an adopted optimal value and mapped through a sigmoid to suppress outliers. The total score was defined as

$$Q = \prod_i f_i \quad (11)$$

where i indicates each category. Thus Q is maximised when images are simultaneously good in all categories.

After visual inspection of the top ranked frames to remove defective or rotated images, 11 images were selected and averaged to build the reference image.

3.2. Determining Free Parameters

The background polynomial degree N_B was first optimised independently, as it is only weakly correlated with the kernel parameters. For 12 representative images (distinct from those used in the reference), DIA was run with $N_B = 0-8$. For each difference image, the mean μ_i and scatter were obtained, and the loss

function

$$L = \frac{1}{N} \sum_i^N \left(\mu_i^2 + \left(\frac{s_i}{5} \right)^2 \right) \quad (12)$$

with $N = 12$, was used to select the optimal N_B . The factor 5 was used to roughly balance the weight of the scatter relative to the mean.

The remaining parameters $(\mathcal{K}, N_n, \sigma_n)$ are strongly correlated, so they were optimised using the `scipy.optimize.dual_annealing` global optimisation algorithm, minimising the loss function in Equation (12) over the same set of 12 images [7]. The best-fitting set of parameters was

$$N_B = 4, \quad \mathcal{K} = 2, \quad N_n = \{1, 2\}, \quad \sigma_n = \{4.74, 1.36\}.$$

These values were then used to generate the full time series of difference images.

4. RESULTS

Light curves at candidate microlensing locations were extracted. Four representative events were selected for detailed analysis. Where appropriate, Equation (4) was fitted to the light curves to obtain $t_{1/2}$ and hence an estimate of the lens mass via Equation (5).

4.1. Event 1

Figure 1 shows the data and fitted curve of Event 1. The light curve exhibits a single, symmetric, non-

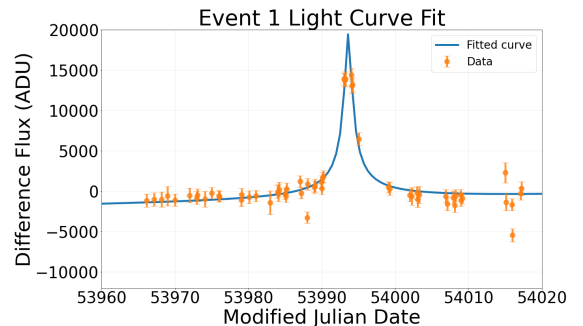


FIG. 1. Event 1: difference-flux light curve and best-fitting pixel-lensing model. Reduced $\chi^2_r = 1.8$, $t_{1/2} = 1.70 \pm 0.24$ days.

periodic peak, consistent with microlensing. A sinusoidal term was added to Equation (4) to describe low-level baseline variability to model the lensing event.

The fit gives $t_{1/2} = 1.70 \pm 0.24$ days. Assuming bulge–bulge lensing with lens–source separation (2 ± 1) kpc and relative transverse velocity $v_{\text{rel}} = (130 \pm 50)$ km s $^{-1}$ Equation (5) yields

$$M_{\text{lens}} = (8 \pm 6) \times 10^{-3} M_{\odot},$$

consistent with a sub-stellar object such as a brown dwarf or massive exoplanet [8, 9].

4.2. Event 2

Event 2 (Fig. 2) shows two distinct peaks separated by ~ 45 days. Fitting each peak independently with Equation (4) and interpreting them as separate microlensing events would imply an extremely low probability about $\sim 10^{-5}$. This suggests it's more likely to be a high-magnification binary lens [10].

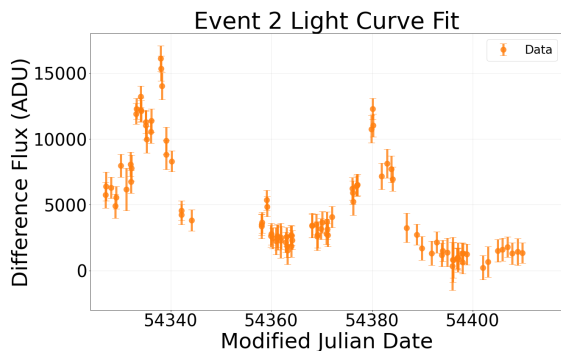


FIG. 2. Event 2: two distinct peaks within ~ 45 days. No single pixel-lensing model provides a good fit.

4.3. Event 3

Event 3 (Fig. 3) shows a sharp rise followed by a slow decay, with multiple peaks. The strong asymmetry

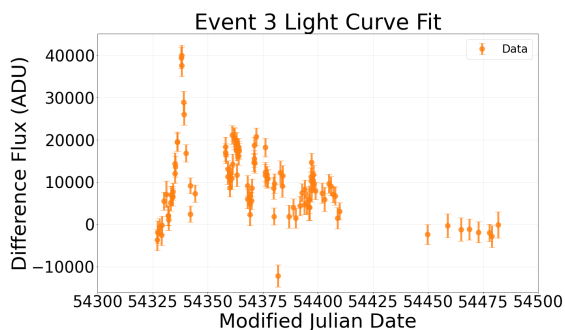


FIG. 3. Event 3: asymmetric peak with a fast rise and slower decay, and multiple sub-peaks. No microlensing fit is shown.

and multiple peaks are not consistent with the PSPL model, but are typical of nova-like transients. Considering the bulge's old stellar population and the relative flux change, a classical nova is a plausible explanation.

4.4. Event 4

Event 4 (Fig. 4) displays a broad, symmetric, non-periodic peak and can be reasonably fitted by Equation (4), giving $M_{\text{lens}} = (5 \pm 4) M_{\odot}$ compatible with a massive star or stellar remnant. However, the old

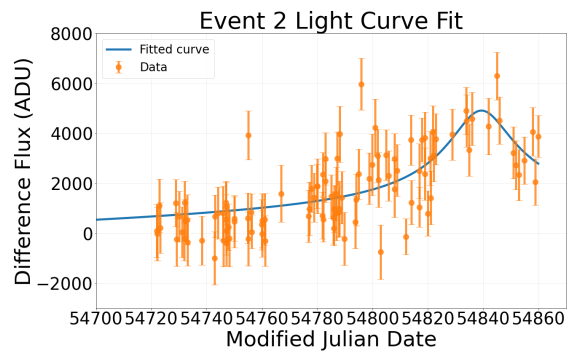


FIG. 4. Event 4: broad, symmetric peak. The fit is plotted, but the microlensing nature is not uniquely established. Reduced $\chi_r^2 = 2.6$, $t_{1/2} = 40.6 \pm 9.6$ days.

bulge population makes a $5 M_{\odot}$ main-sequence star unlikely. The event could arise from a compact remnant (white dwarf, neutron star or black hole), but alternative explanations exist. For example, a long-period variable such as a Mira can produce a single apparent peak within a ~ 3 yr monitoring window [11]. In addition, A bulge-disk geometry with lens-source separation (10 ± 5) kpc and $v_{\text{rel}} \approx 200 \pm 70 \text{ km s}^{-1}$ gives $M_{\text{lens}} \approx 2.3 \pm 1.8 M_{\odot}$, which is also plausible. This degeneracy cannot be broken with a single-band light curve; multi-band photometry would be required to distinguish between microlensing and intrinsic variability.

5. CONCLUSION

Difference imaging analysis was applied to Angstrom survey data to identify microlensing candidates in the M31 bulge. A reference image and kernel-background model were constructed using optimised parameters, allowing reliable extraction of difference flux light curves. Four representative events were examined: one showed a symmetric peak consistent with microlensing by a sub-stellar lens; one was better explained by a binary-lens configuration; one displayed asymmetric, multi-peaked behaviour typical of a nova; and one broad symmetric event was compatible with microlensing but also consistent with long-period stellar variability. These results demonstrate both the capability of DIA to recover microlensing signals in crowded fields and the degeneracies that limit mass determination without multi-band data.

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